

PRACTICE MODULE BRIEFING for Certificate in Intelligent Reasoning Systems (IRS)

Machine Reasoning (MR)
Reasoning Systems (RS)
Cognitive Systems (CGS)

by Dr. Gary Leung (cc.leung@nus.edu.sg)

Certificate in Intelligent Reasoning Systems (IRS)

Goal: Teach participants the knowledge, skills and industry best practices to design and build Intelligent Systems that solve problems by computational reasoning using captured domain knowledge as well as knowledge discovered/learnt from both small and big data.

Key Takeaways:

- Expose participants in a supervised manner to real world problems so that they may practice the use of the skills they have learned during the individual course modules in a real-world setting and obtain expert advice and guidance when needed.
- Enable participants to demonstrate their proficiency across all of the skills that they have learned in the course modules and hence be certified as competent at the Certificate level.
- Provide a formal grading mechanism so that the certificate may be used as one component in the NUS-ISS Stackable Master of Technology (MTech) in Artificial Intelligent Systems.

Agendas

Machine Reasoning

Day 1

- Machine Reasoning Overview
- Reasoning as Search
- Workshop: Search for Problem Solving

Day 2

- Reasoning for Problem Solving
- Knowledge Representation
- Workshop: Reasoning using Different Techniques
- Representation of Models from Machine Learning

Day 3

- Machine Reasoning over Knowledge Graphs
- Reasoning Methods over Knowledge Graphs
- Knowledge Graph Embeddings
- Similarity-based Recommender Systems & Workshop

Day 4

- Inductive Reasoning and Learning
- Neural Network Learning
- Workshop: Machine Learning Tasks & Models

Reasoning Systems

Day 1

- Reasoning for Problem Solving
- Modern Reasoning Architectures
- Further Discussion

Day 2

- Reasoning and Knowledge Discovery from Large Datasets
- Market Based Analysis and Recommender Systems & Workshop
- Similarity-based Recommender Systems & Workshop

Day 3

- Model-based Recommender Systems & Workshop
- Hybrid and Advanced Recommender Systems
- Hybrid Recommender Systems Workshop

Day 4

- Reasoning using Optimization Techniques
- Evolutionary Computation and Genetic Algorithm Applications
- Workshop: Genetic Algorithms

Day 5

- Advanced Embedding Techniques
- Knowledge Graphs Workshop

Cognitive Systems

Dr. FAN Zhenzhen



• Day 1 (AM)

- Introduction of Cognitive Systems
- Natural language cognition

Dr. CHENG Jierong



• Day 1 (PM)

- Vision Cognition

Dr. Gary LEUNG



• Day 2

- Audio cognition

Dr. WANG Aobo



• Day 3

- Knowledge representation and reasoning
- Case studies and workshops

Graduate Cert: Assessment Components

- The graduate cert assessment comprises the assessment components below:

Assessment Component	Weight
Examination	50%
Practice Module Project Work	50%

Standard Grading Scheme		CAP
A+	85 - 100	5.0
A	80 - 84	5.0
A-	75 - 79	4.5
B+	70 - 74	4.0
B	65 - 69	3.5
B-	60 - 64	3.0
C+	55 - 59	2.5
C	50 - 54	2.0
D+	45 - 49	1.5
D	40 - 44	1.0
F (Fail)	0 - 39	0.0

- A participant must attain a minimum overall score of **50%** to pass the practice module and hence be awarded the Certificate in Intelligent Reasoning Systems.
- Note:** The participant must pass ***both written examination*** and ***Practice Module***.

Graduate Certificate: page 1 of 13

Name	:
Email	:
Phone No.	:
NUS Matriculation No. (If applicable)	:

Institute of Systems Science
National University of Singapore

GRADUATE CERTIFICATE
INTELLIGENT REASONING SYSTEMS

Exam

Subject: _____

SECTION A

Question	Marks
1	
2	
TOTAL	

Instructions for Paper

Date:
Time:
Duration:
Place:

This is an OPEN BOOK examination. This examination paper consists of one Section and two Questions. You are to answer ALL questions. There are a total of Marks for this paper.

Version 2018 09 19



- **3-hour open-book individual exam in the 1st-2nd week of Nov 2026 (tentative).**
- **Pencil can be used for drawing.**
- **Internet (re)search is NOT allowed.**
- **Bring your NUS matriculation ID card and Identification Card.**

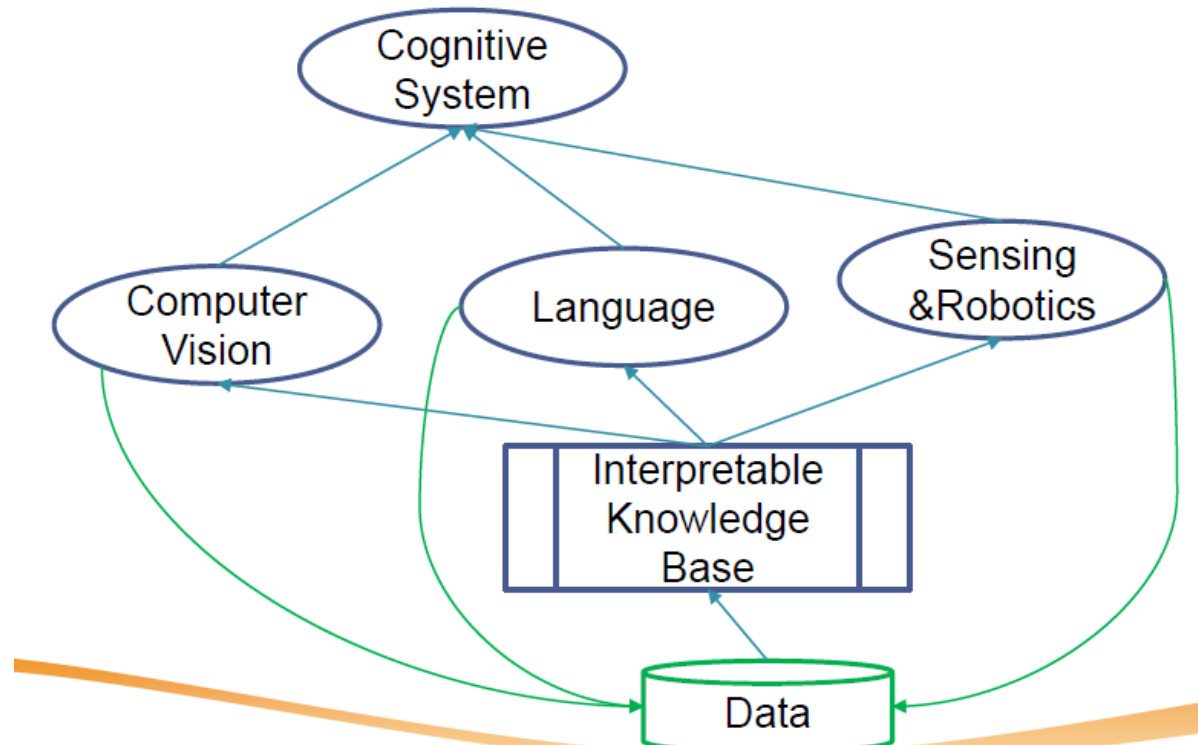
[Group Project] What are the requirements?

- Form a project team of max **5** members, appoint a team leader.
- Identify a relevant business/research problem in the domain of **knowledge discovery, hybrid machine reasoning, recommendation or optimization**. Propose, design and create an **intelligent reasoning** system for the business need/problem.
- Projects may be selected from within their own organizations or from other approved sources. These projects will be assessed by ISS lecturers.
- Practice module takes an estimated **10 man-days** of effort by each participant.
- The proposed project must develop, integrate, and demonstrate any **three** or more aspects out of following **four** technique groups:
 - Decision automation: Business rules & process **OR** Knowledge based reasoning techniques
 - Business resource optimization: Informed search **OR** Evolutionary computing techniques
 - Knowledge discovery & (big) data mining techniques, e.g. recommendation; diagnosis; etc.
 - System designed with cognitive techniques or tools, e.g. components of knowledge base: knowledge graph, frame systems, user-interface supporting human mode of communication like natural language, chat-bot, etc.

To minimize “reinventing the wheels”, the delivered system is encouraged to adopt existing **open source sub-modules and/or application programming interfaces APIs**. Teams can explore other relevant technologies to extend the system/solution for more effective business problem solving, which can be considered for **value-added** marks. Teams must demonstrate (**write in report**) their ability to apply and practice the knowledge, techniques and skills they have learnt from the courses.

What are Cognitive Systems?

- Systems that exhibit **human-like** intelligence through processes like **perception, learning, reasoning, and memory.**



Cognitive Systems Application Areas

- **Product** applications embed cognitive technologies into a product or service to deliver customer benefits such as ease of use, simplicity, and automation.
- **Process** applications embed these technologies into an organization's workflow, automating tasks to execute operations faster, more cost-effectively, and at a higher quality.
- **Insight** applications leverage analytic capabilities and machine learning to uncover deep insights, enabling better data-driven operational and strategic decisions.

[Group Project] What to develop?

- A 5-minute video clip to promote/sell your system/solution, covering:
 - Business Pain & Value; Use Case Demo
- A 5-minute video clip to describe high level system design your system/solution, covering:
 - System Design; Technical explanation of use cases
- A **runnable** intelligent reasoning system;
- A **group project report** (.doc or .pdf) with relevant attachments, including:
 - Business Case / Market Research / Literature Review
 - System Design / Model
 - System Development & Implementation
 - Experiments
 - Findings and discussion
- Appendix of **report**: Project Proposal
- Appendix of **report**: Mapped System Functionalities against knowledge, techniques and skills of modular courses: MR, RS, CGS
- Appendix of **report**: Installation and User Guide
- 1 pages of **individual project report** per **project member**, including:
 - Individual reflection
 - Peer review form

Practice Module

[Group Project] What to submit?

Create Github repository using template:

<https://github.com/IRS-PM/Workshop-Project-Submission-Template>

As a project group: Prepare below **four** files with naming convention:

1. A **member-github.txt**, containing:
 - Group name, e.g. AwsomeSG
 - All member's full name (as shown on your student ID/NRIC/FIN/Passport)
 - All member's student IDs, e.g. A1234567B; or masked NRIC, e.g. S****123T
 - Weblink to your online github project repository, e.g. <https://github.com/>...
2. Two separate video files, e.g. **IRS-PM-2021-01-16-IS03PT-GRP-AwsomeSG-ReleaseOptimizer-promotion/system.mp4/wmv/mov/avi/etc**
3. A downloaded github repo zip, e.g. **IRS-PM-2021-01-16-IS03PT-GRP-AwsomeSG-ReleaseOptimizer.zip**
4. A group project report, e.g. **IRS-PM-2021-01-16-IS03PT-GRP-AwsomeSG-ReleaseOptimizer-Group-Report.pdf**

Finally, compress all above files into one single .zip file, naming convention, e.g. **IRS-PM-2021-01-16-IS03PT-GRP-AwsomeSG-ReleaseOptimizer.zip**, then upload to **Canvas Assignments: [Group Leader] Project Deliverables (zip)** folder.

As an individual: Prepare and submit confidential **Individual reflection** to **Canvas Assignments: [Individual] Project Peer Evaluation (doc, docx, pdf)** folder.

Practice Module

[Group Project] What to submit?

Canvas Assignments:

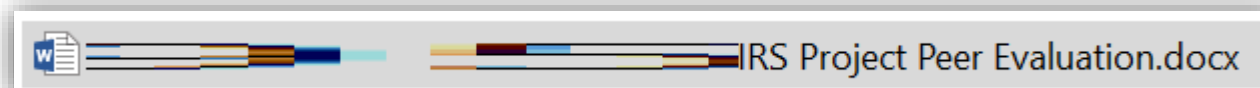
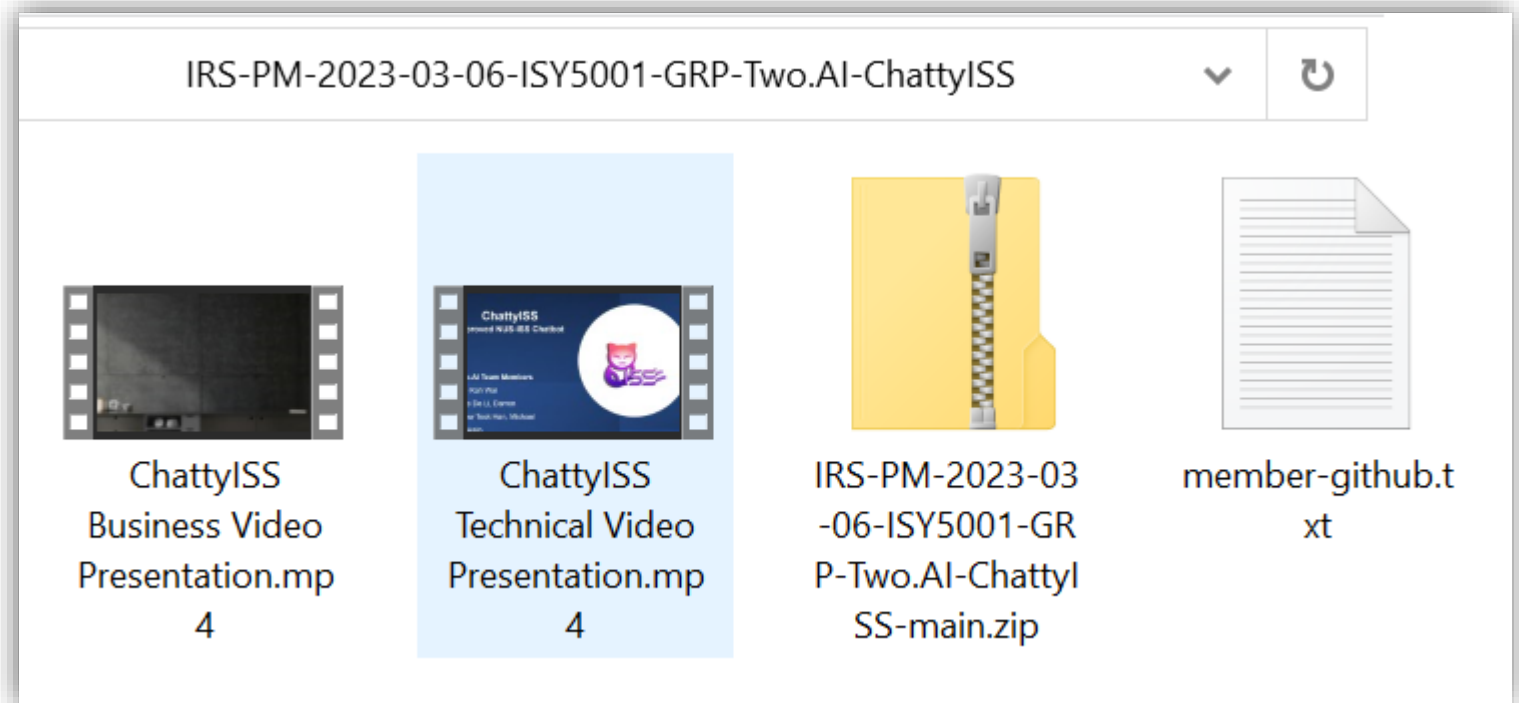
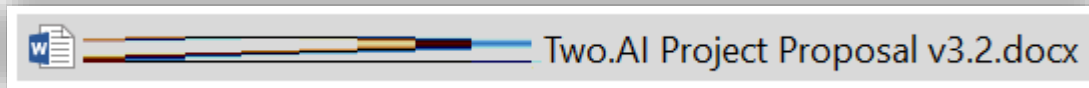
*[Group Leader] Project Proposal
(doc, docx, pdf) folder.*

Canvas Assignments:

*[Group Leader] Project
Deliverables (zip) folder.*

Canvas Assignments:

*[Individual] Project Peer
Evaluation (doc, docx, pdf)
folder.*



[Group Project] Where to submit?

Welcome!

Announcements

NUS students: change of login mechanism
20 Dec 2022

Join the geNiUSchannel & geNiUSbooks 2022 competition to win exciting prizes!!!
06 Oct 2022

Canvas course sites for AY2022–2023, Sem 2
14 Sep 2022

Transitioning from LumiNUS to Canvas
15 Aug 2022

Guides

 Staff

 Student

Support

 Email IT Care

 Call IT Care



Browse
Canvas Courses

Browse
NUSMods

Sign In

Login to your account.

NUS Students / Alumni

NUS Staff / Visitors

Or

Non-NUS Users

[Group Project] How to evaluate?

The deliverables will be assessed in a scheme including aspects of:

1. **Business Value** (including market research or literature review; degree of novelty)
2. **System Design**, e.g. smart functions and technological advancement (shown in report/ documentation; degree of novelty)
3. **System Implementation** (system functionality and usability; demonstrated through the system and user guide)
4. **Experiments** (quantitative evaluation of proposed methods)
5. **Presentation Videos** (AI-generated presentation speech is not allowed)
6. **Project Contribution**: peer-review; team-size
7. **Value Adds**, e.g. relevant techniques learned via self research; intuitive user interface

[Submission deadline]

- Project proposal due by 23:59 on 13rd Sept 2026 (Sun.)
- Proposal presentation on the following week (TBA)
- Submission of final project deliverables and individual peer review due by 23:59 on 25th Oct 2026 (Sun.)

[Group Project] What are some project ideas?

- Children Development Monitor / Milestone Tracker <https://www.cdc.gov/ncbddd/actearly/index.html>
- Augmented Reality: Evidence-based tools to help you teach social-emotional skills to students with autism or ADHD <http://www.brain-power.com/>
- Dynamic Book Order Handling System https://github.com/IRS-MR/IRS-MR-2019-01-19-IS1PT-GRP-X-Men-Online_Order_Management_System
- Query Answering Information Retrieval Chat-Bot <https://github.com/IRS-CGS/IRS-CS-2019-04-27-IS1PT-GRP-ISSChatBot>
- Algorithmic Trading System (ATS) for Crude Palm Oil Futures <https://github.com/IRS-PM/Workshop-Project-Submission-Template-Trading>
- IT Project Release Optimizer <https://github.com/IRS-PM/Workshop-Project-Submission-Template-GA-Optimizer>
- IBM Watson® Discovery service <https://www.ibm.com/cloud/garage/architectures/cognitiveDiscoveryDomain/overview>
- IBM Watson® Assistant <https://www.ibm.com/cloud/watson-assistant/>
- Google Dialogflow <https://dialogflow.com/>
- Mycroft <https://mycroft.ai/>

Past student project references: <https://github.com/IRS-PM>

Practice Module : MVP Project

[Example Past Project]

<https://github.com/IRS-PM>

[Home Improvement] Cooking Assistant

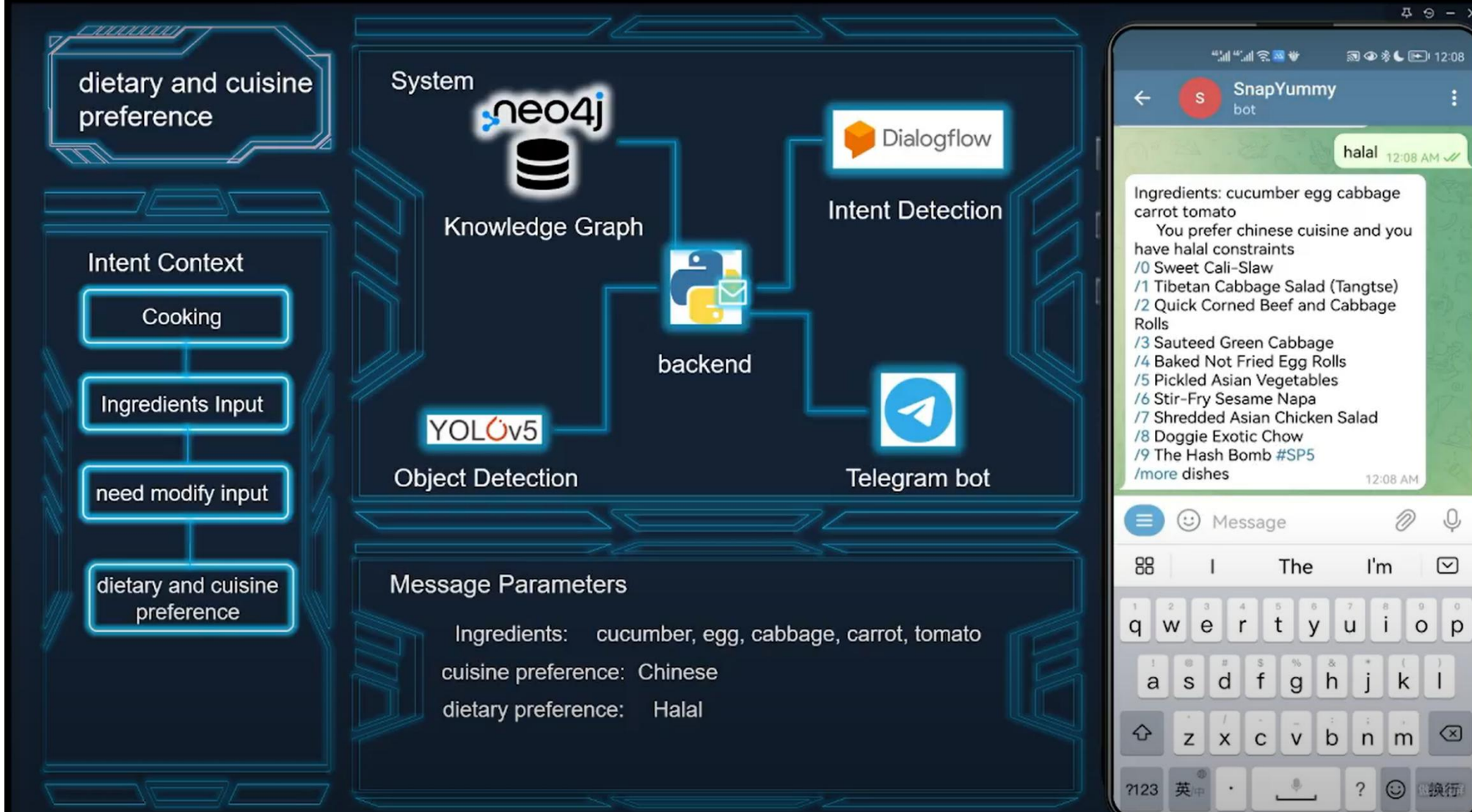
SnapYummy

An Intelligent Cooking Assistant at your fingertip

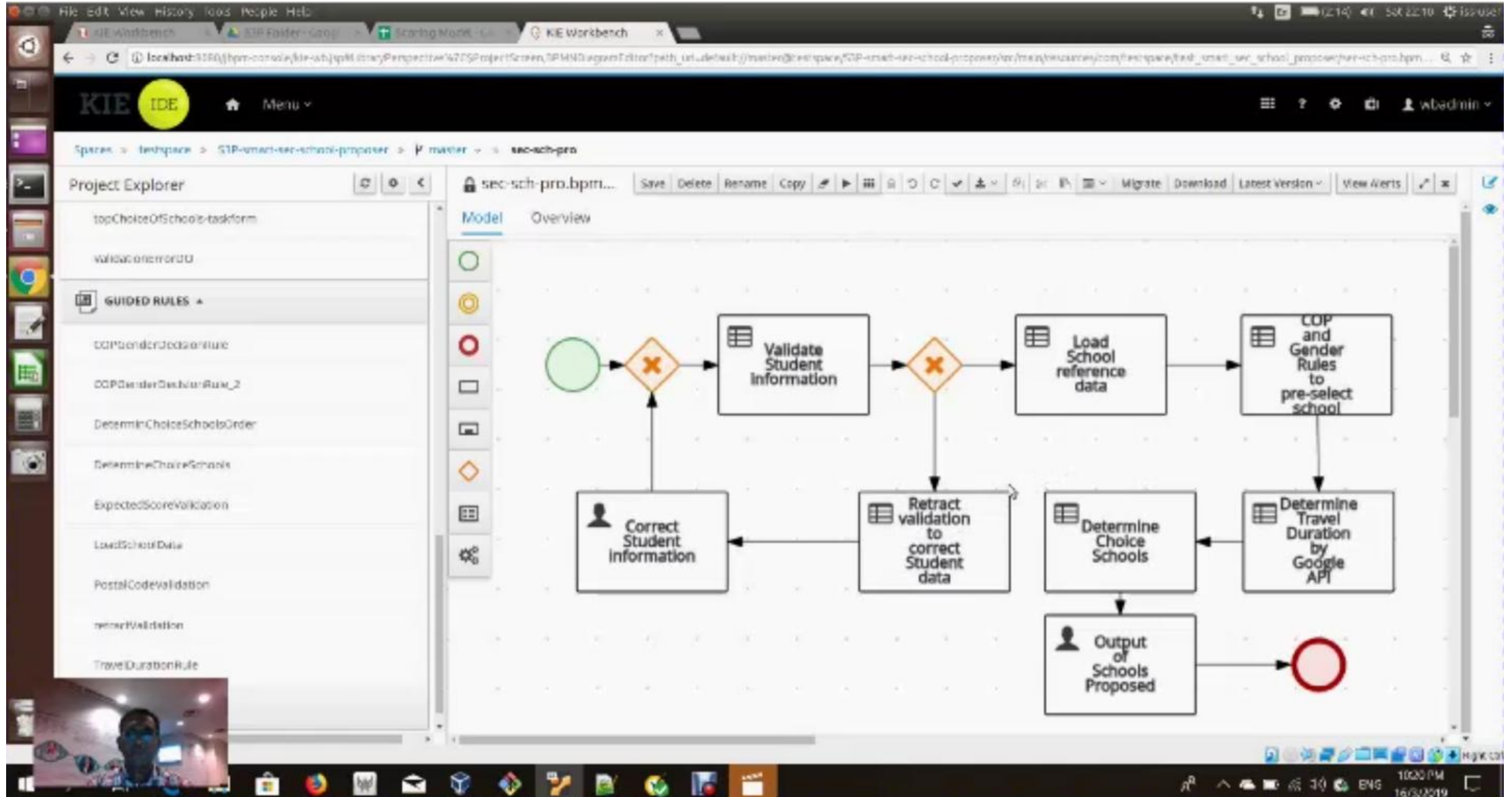


[Home Improvement] Cooking Assistant

IRS-Group 8.mp4



[Education] Secondary School Proposer



[Education] School Shuttle Scheduler

Intelligent Rapid Shuttle

AI powered Shuttle Service -The fastest way to reach School at an affordable price

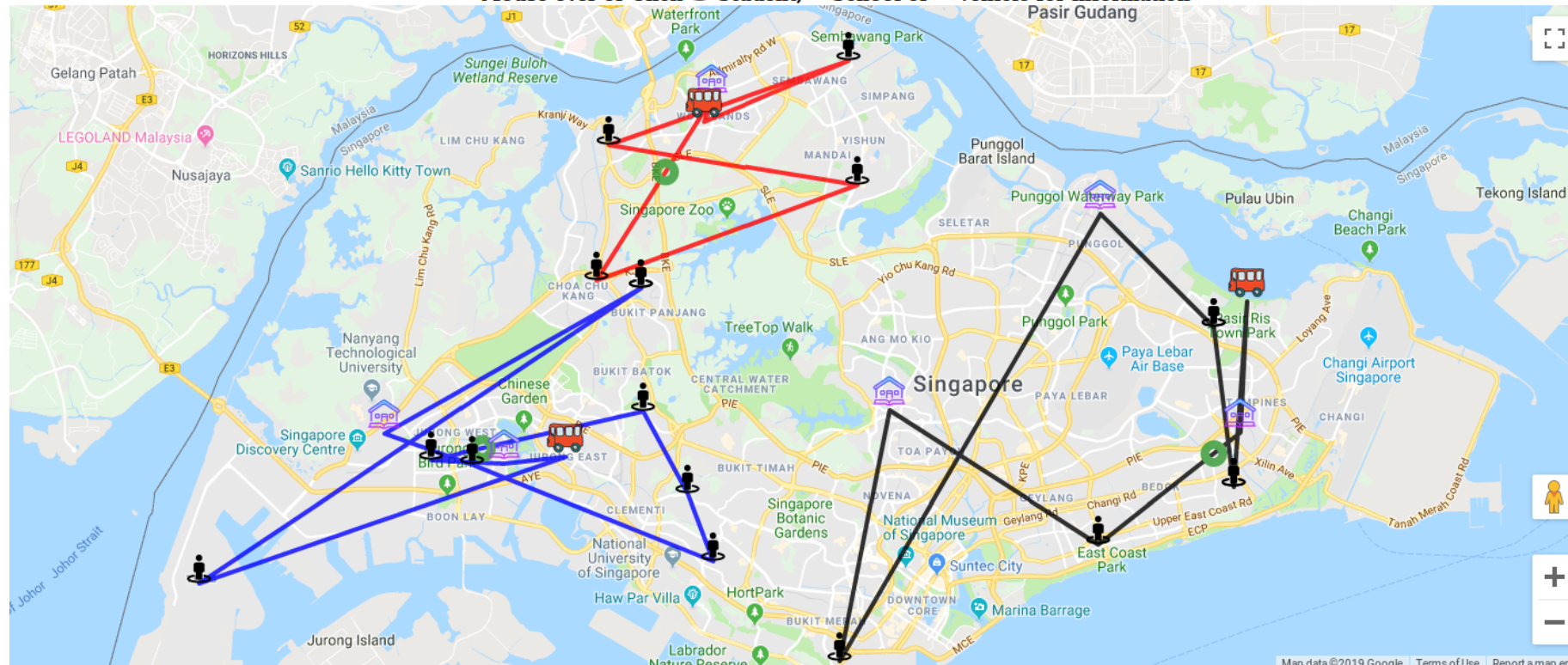
- ☐ DataSet 1(8 Students, 3 Schools, 2 Vehicles)
- ☒ DataSet 2(18 Students, 5 Schools, 3 Vehicles)
- ☐ DataSet 3(23 Students, 8 Schools, 3 Vehicles)
- ☐ DataSet 4(Same as Dataset3 + 1 Vehicle)

Show on Map

Find Best Route for the Students

Score: 0hard/-3381soft

Mouse over or Click  Student,  School or  Vehicle for information



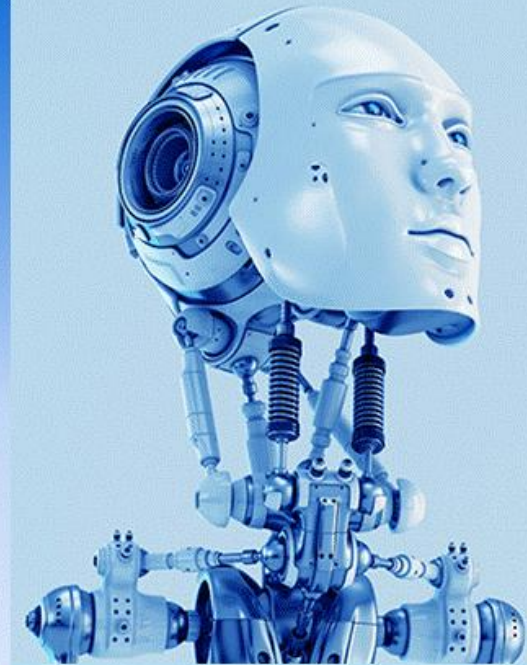
[Health] Depression Screener



Pepper Project Group

GAO LIANG	A012884E
GENG LIANGYU	A0195278M
HAN DONGCHOU RANCIS	A0195414A
ONG DON PING	A0195172B
TAN CHIN GEE	A0195296M

Depression Screening System
Depression Screening System

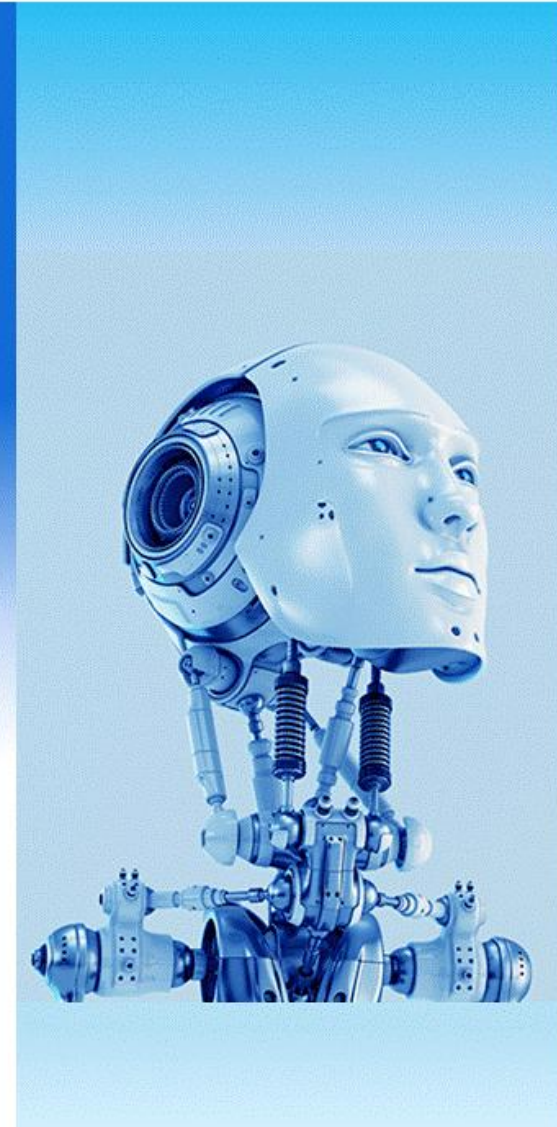
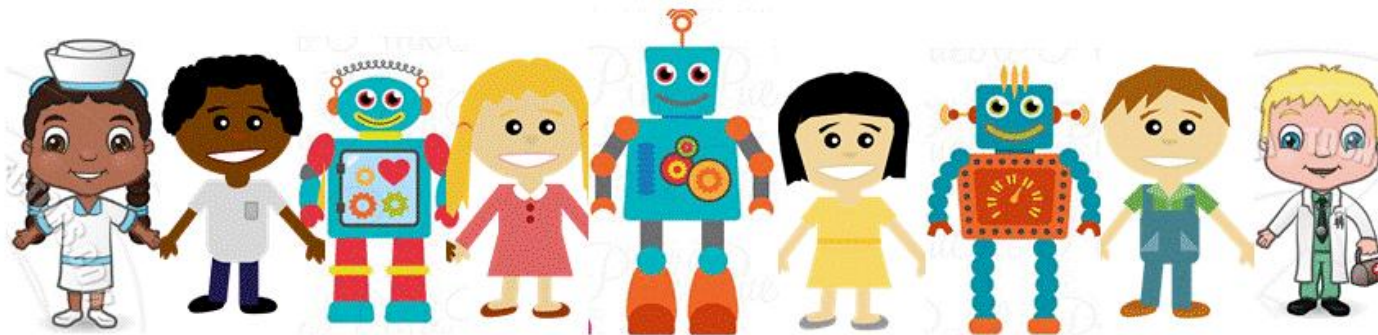


[Health] Patient-Doctor Matcher

Pepper Project Group

CAO LIANG	A0012884E
GENG LIANGYU	A0195278M
HAN DONGCHOU FRANCIS	A0195414A
ONG BOON PING	A0195172B
TAN CHIN GEE	A0195296M

Patient Matching System



[Manufacturing] Integrated Circuits Lot Disposition Recommender



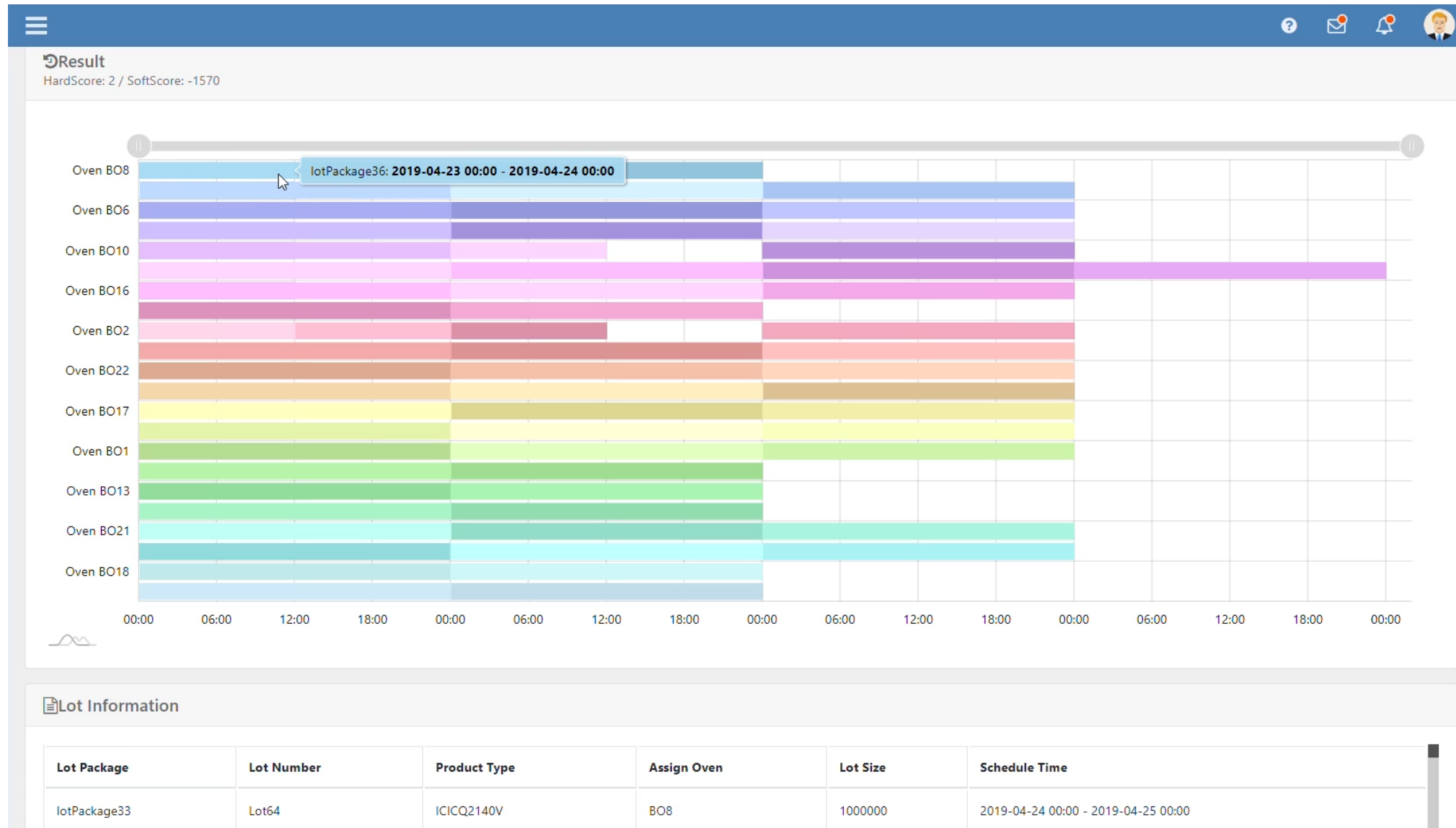
LOT DISPOSITION RECOMMENDER SYSTEM

Username

Password

Sign In

[Manufacturing] Integrated Circuits Lot-Oven Scheduling and Dispatch Optimizer



[Health] Meal Planner for Diabetics



[Education] ISS Course Recommender



NICF- Intelligent Sensing and Sense Making (SF)	
Class Name	Class 1
Class Time	9:00am - 5:00pm
Start Date	2019-11-25
End Date	2019-11-28

NICF- Pattern Recognition and Machine Learning Systems (SF)	
Class Name	Class 1
Class Time	9:00am - 5:00pm
Start Date	2020-01-06
End Date	2020-01-10

NICF- Problem Solving using Pattern Recognition (SF)	
Class Name	Class 1
Class Time	9:00am - 5:00pm
Start Date	2019-11-04
End Date	2019-11-21

Merchant Onboarding

Automated Screening Service Presentation.pptx - PowerPoint



Trustworthiness Validation Solution

Cloud-ready | Extensible | Fast | Saves Cost

[FinTech] Co-branded Petrol Credit Card

FIND ME A PETROL STATION ALONG THE WAY

CHOOSE CREDIT CARD

to look for discounts on the available petrol station brands



SELECT YOUR DESTINATIONS

Please enter valid postal codes. You can indicate up to 10 destinations.

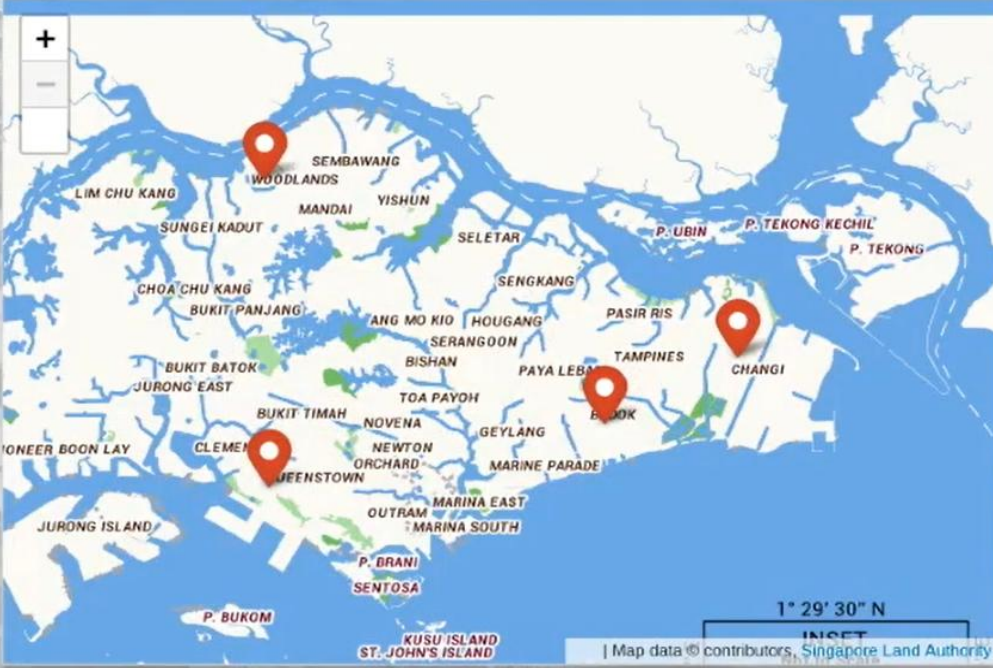
Added Markers 4/10

1

21 LOWER KENT
RIDGE ROAD
NATIONAL
UNIVERSITY OF
SINGAPORE
(LT20)
SINGAPORE
119077

2

137 MARSILING
ROAD HDB-
WOODLANDS
SINGAPORE
730137



1° 29' 30" N

Map data © contributors, Singapore Land Authority

Except the first(start) and last(end) points, the middle points can be fixed or otherwise. Please select at least 2 points.

My route is fixed

No fixed order

© 2026 National University of Singapore

28

[E-Commerce] Shipping and Packing Optimizer



Pro Store

Sign up

Sign in

 Cart



Toy Car
\$100.00



Cotton Mat
\$19.00



Chair
\$299.00



Black Watch
\$50.00



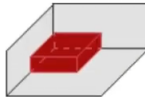
Brown Watch
\$50.00

Delivery Name	Delivery Time (Days)	Shipping Cost	Select
 SF	2 to 3	\$35.5	☐
 UPS	3 to 5	\$77.4	☐
 DHL	2	\$72.75	☐
 FedEx	4	\$72.6	☐

Note Your package will be packed in the following manner to minimize shipping cost!



30x15x30 **Final Packing**



15x5x15 **Toy Car**

THE END